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Spaceflight-induced contractile and mitochondrial dysfunction in an automated heart-on-a-chip platform

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With current plans for manned missions to Mars and beyond, the need to better understand, prevent, and counteract the harmful effects of long-duration spaceflight on the body is becoming increasingly important. In this study, an automated heart-on-a-chip platform was flown to the International Space Station on a 1-mo mission during which contractile cardiac function was monitored in real-time. Upon return to Earth, engineered human heart tissues (EHTs) were further analyzed with ultrastructural imaging and RNA sequencing to investigate the impact of prolonged microgravity on cardiomyocyte function and health. Spaceflight EHTs exhibited significantly reduced twitch forces, increased incidences of arrhythmias, and increased signs of sarcomere disruption and mitochondrial damage. Transcriptomic analyses showed an up-regulation of genes and pathways associated with metabolic disorders, heart failure, oxidative stress, and inflammation, while genes related to contractility and calcium signaling showed significant down-regulation. Finally, in silico modeling revealed a potential link between oxidative stress and mitochondrial dysfunction that corresponded with RNA sequencing results. This represents an in vitro model to faithfully reproduce the adverse effects of spaceflight on three-dimensional (3D)-engineered heart tissue.

spaceflight | heart-on-a-chip | mitochondrial dysfunction | microgravity | oxidative stress

Significance

Our study demonstrates the adverse effects of spaceflight on an automated heart-on-a-chip platform sent to the International Space Station. This platform replicates the intricate environment of the human myocardium, offering a truer depiction of cardiac behavior beyond conventional two-dimensional (2D) cell cultures and animal studies. The innovation is underscored by the design of a device that requires minimal handling by the astronauts to maintain viability in Space. Central to the observed tissue dysfunction are mitochondrial impairment and potential oxidative stress, manifesting as increased arrhythmias during space missions and a sustained loss of contraction strength, persisting even after returning to Earth's gravity. These spaceflight-induced changes mimic cardiac aging, underscoring the model's potential use in aging-related cardiovascular disease research.

Since the space race of the 1960s, over 550 humans have flown into space. The establishment of the International Space Station (ISS) has allowed these forays to last for increasingly longer durations. While long-duration (>300 d) missions are still rare, planned missions to Mars and beyond represent an urgent need for an improved understanding of how the harsh environment of space impacts the human body. Much is already known about the detrimental impact of spaceflight on human health, including loss of muscle mass (1), bone density (2), and cardiovascular function (3, 4). The effects on the cardiovascular system remain of particular concern, as even short-term cardiovascular failure can have devastating effects. In fact, the Apollo astronauts who traveled to the moon were nearly 5 times more likely to die from cardiovascular disease than astronauts who never left low Earth orbit (5). Within 24 h of leaving Earth's gravitational forces, astronauts experience increased ventricular volumes and decreased central venous pressure (3). Furthermore, a seminal study known as the NASA Twins Study found that an astronaut who spent 340 d on the ISS exhibited significantly increased cardiac output and reduced arterial pressure compared to an Earth-bound identical twin, likely due to decreases in afterload (6), or the amount of work the heart has to do to pump blood. These changes could lead to adverse cardiac remodeling and deconditioning and contribute to the observed orthostatic intolerance in astronauts returning from spaceflight (7). Arrhythmic events, including atrial fibrillation and ventricular tachycardia, have occurred in astronauts during spaceflight, despite preflight screening indicating no preexisting cardiovascular disease in the same astronauts (8). Although many of these changes remained stable or reverted to baseline after return to Earth, the implications of how these risk factors would impact longer-duration missions are unclear.

Decades of research have started to untangle the molecular determinants of this cardiovascular dysfunction, primarily using animal models and nonhuman cardiomyocytes. Studies have identified spaceflight-induced alterations in the transcriptional expression of rat cardiac myosin (9) and mitochondrial genes (10), indicating alterations in the contractile apparatus and metabolic function of cardiomyocytes. Another study found increased localization of nuclear transcription factor κB (NF-κB), which is implicated in muscle atrophy and plays a key role in regulating the immune response to oxidative stress (11) in rat cardiomyocytes cultured in simulated microgravity (12). Despite this research, we are still far from identifying the key pathways involved in human cardiomyocyte responses to microgravity and developing

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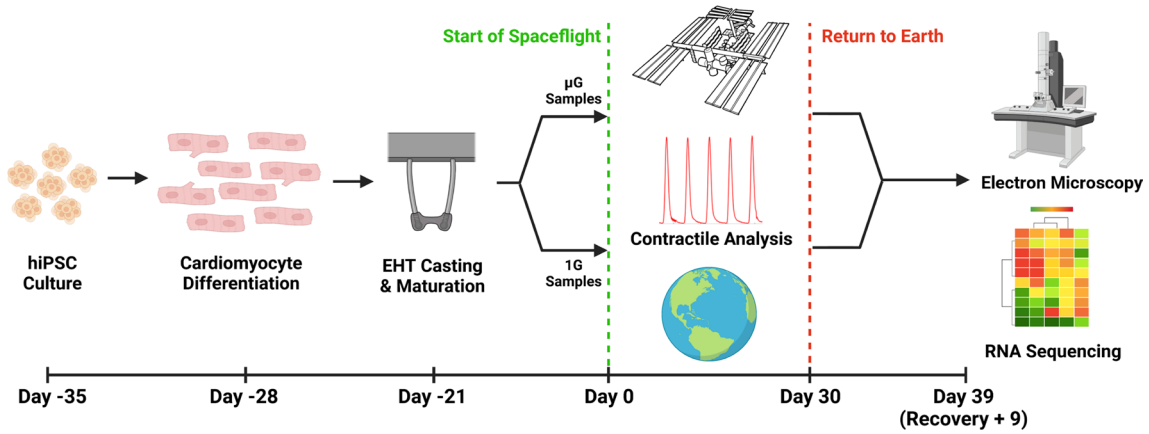


Fig. 1. Schematic illustration of EHT fabrication and experimental timelines. hiPSCs were differentiated into cardiomyocytes which were then used to fabricate EHTs at 20 d postdifferentiation. EHTs were loaded into chambers and maintained for 3 wk before launch. Tissues remained on board the ISS for approximately 4 wk before being returned to Earth, at which time a subset of tissues were maintained for an additional week. RNA sequencing and TEM were performed at the conclusion of the experiment. Spaceflight and 1G ground control experiments were conducted with identical timelines.

methods to mitigate its damage, particularly since interspecies differences in cardiovascular biology can impair the translatability of these findings (13). More studies in a human context are required; however, astronauts are too few and too valuable to rapidly investigate potential interventions safely and effectively. Therefore, representative in vitro models of human cardiac tissue must be developed that can be deployed to the ISS as experimental platforms.

The emergence and adoption of human-induced pluripotent stem cells (hiPSCs) as a research tool has improved the methods used to model human organ systems and disease pathologies by circumventing issues with primary cell tractability and the use of nonhuman cells and animal models. This has led to a study in which hiPSC-derived cardiomyocytes (hiPSC-CMs) were sent to the ISS to study the effects of spaceflight on human cardiac function. In this study, key changes in calcium handling and gene transcription were identified, especially with regard to mitochondrial function (14). However, the reliance of many of these tissue models on two-dimensional (2D) culture platforms is not ideal, as it has been frequently observed that cells cultured in 2D exhibit significant differences in structure, function, and gene expression compared to their in vivo or three-dimensional (3D) counterparts (15). This is particularly true of hiPSC-CMs, which are phenotypically immature immediately postdifferentiation and fail to demonstrate appropriate transcriptional and functional development when cultured on stiff 2D surfaces (16). Due to this neonatal-like phenotype, results from studies involving hiPSC-CMs are not translatable nor fully comparable to studies in adult humans, and in the context of space biology, this may lead to difficulties in using these cells for understanding and clarifying the underlying causes of microgravity-induced heart failure. These issues have led to the development of 3D platforms such as engineered heart tissues (EHTs), in which hiPSC-CMs are cultured within a hydrogel scaffold material to produce more representative in vitro models capable of better reproducing the impact of external stimuli on human cells and organs (16–18). hiPSC-CM maturation in 3D can be further improved through advances in EHT scaffold technology. For example, our group has demonstrated that the incorporation of an electroconductive synthetic material like reduced graphene oxide (rGO) in decellularized myocardial extracellular matrix (dECM) hydrogels produced tissues with significant enhancements in contractile and electrophysiological function, as well as in adult cardiomyocyte transcriptional marker expression. They were also able to respond appropriately to clinical doses of a known hERG channel blocker (19).

This study employed a heart-on-a-chip platform with dECM-rGO scaffolds for improved tissue maturity and automated contractile function monitoring to evaluate the effects of spaceflight. The EHTs were sent to the ISS, observed for 30 d in microgravity, and subsequently examined during a 9-d recovery period on Earth. These tissues were then compared to an identical set that was cultured on Earth for the same duration and in the same hardware (Fig. 1). Using structural, transcriptomic, and in silico analysis, we identified key changes in calcium handling and gene transcription, especially with regard to mitochondrial function (14) in the microgravity exposed tissues and determined that mitochondrial dysfunction and oxidative stress are key regulators of cardiac dysfunction in spaceflight. These results demonstrated that our experimental approach can be applied to more targeted future studies as well as the screening of potential therapeutic interventions to ameliorate these effects in astronauts.

Results

Development of a Spaceflight-Capable Heart-on-a-Chip Platform with Real-Time Twitch Force Measurements. The method of producing EHTs mounted on posts for the functional assessment of cardiomyocytes has been well established as a powerful tool for in vitro disease modeling and drug discovery since its invention more than 20 y ago (20–22). To date, however, these tissues are often fabricated using a scaffold biomaterial such as fibrin or collagen that, while biological in nature, does not adequately recapitulate many of the microenvironmental cues present in the native myocardium. As previously discussed, a scaffold with the combination of tissue-specific biochemical and electroconductive cues imparted by dECM and rGO, respectively, significantly improves the maturity of EHTs which can then produce clinically relevant responses to external stimuli (19), and it is for this reason that dECM-rGO hybrid scaffolds were used in our study (Fig. 2*A*). In conjunction with this, the tip of the flexible post in our two-post system incorporated a small magnet that allowed for the real-time capture of contractile force, frequency, and beat interval irregularity of EHTs without the use of imaging modalities (23). Specifically, as the tissue contracts, the displacement of the flexible post moves the magnet. The resulting changes in magnetic flux, measured by giant magnetoresistive (GMR) sensors, can be calibrated to the post displacement, and force can be calculated by using Euler–Bernoulli beam bending theory, such that $F = \frac{3EI\delta}{L^3}$, where F is force, E is

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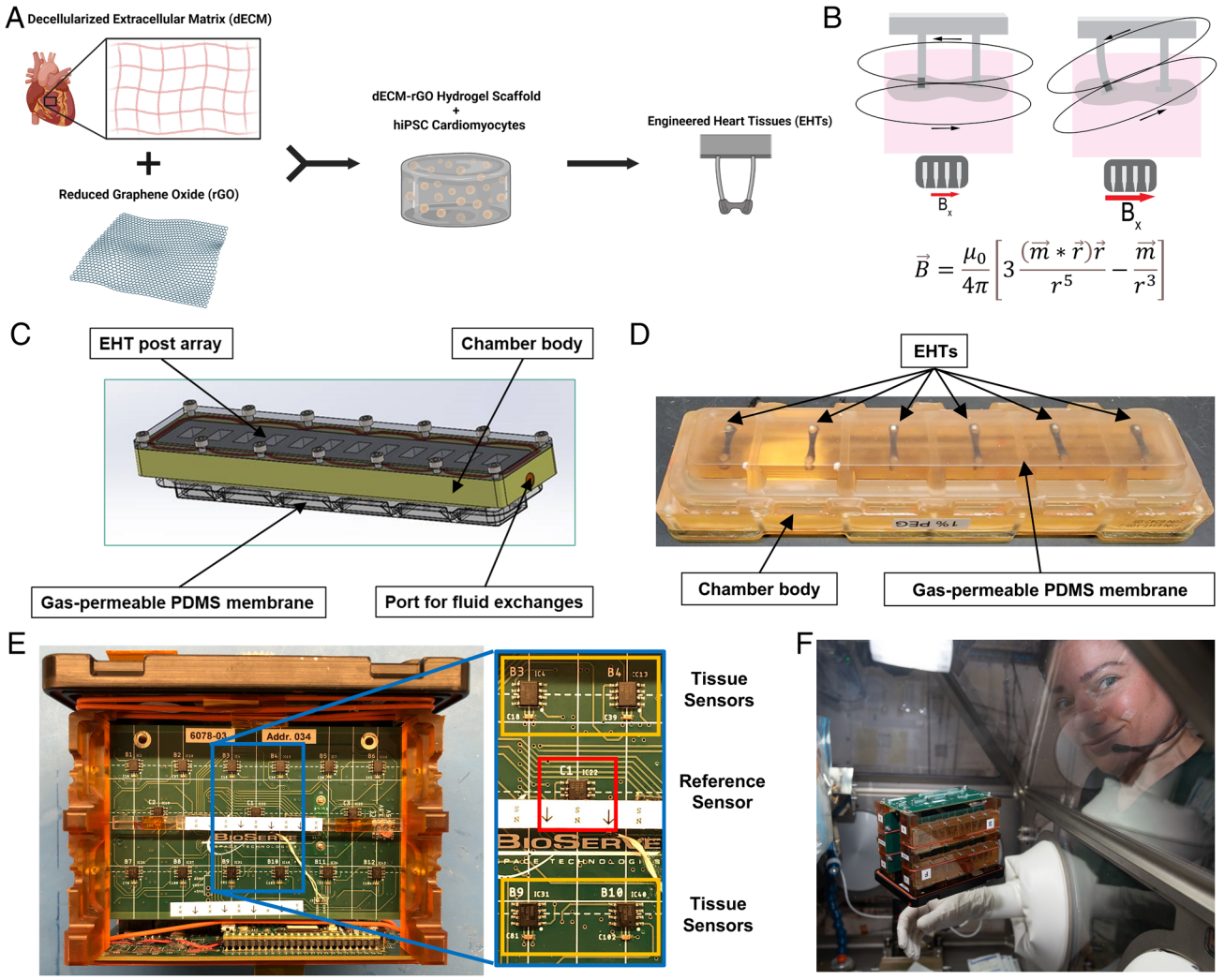


Fig. 2. Hybrid scaffolds and magnetic sensing allow for real-time automated force measurements of tissues with enhanced maturity. (*A*) Scaffolds used to generate EHTs are composed of dECM and rGO. (*B*) Tissue contraction was measured using a series of GMR sensors that convert magnetic moments to force. (*C*) Each EHT array was inserted into a sealed chamber that allowed for fluid and gas exchange. (*D*) Representative photograph of a chamber containing 6 tissues. (*E*) Representative photograph of the printed circuit board within the PHAB containing the GMRs. Using a reference sensor, magnetic noise could be reduced in the final signal. (*F*) Photograph of the experiment during a media exchange conducted by the crew while on board the ISS.

the Young's modulus, I is the moment of inertia of a circle, δ is the deflection of the beam, and L is the height of the post (Fig. 2*B*).

Custom-made tissue chambers were designed and developed to enable the survival and culture of EHTs in the microgravity environment of the ISS, where conventional multiwell culture plates could not be used (Fig. 2 *C* and *D*). These gas-permeable chambers fully encapsulated a single EHT lattice holding six individual tissues and featured several distinctive features that made them suitable for spaceflight. The tissue chambers were then integrated with GMR sensors and associated electronics that were mounted onto printed circuit boards within a plate habitat (PHAB). In addition to a GMR sensor located directly beneath each EHT, three reference GMR sensors were present for each pair of chambers. These reference sensors assist in the elimination of external electromagnetic noise, and when coupled with the known distance between GMR sensors and tissues, readings could be properly calibrated and converted into force measurements (Fig. 2*E*). Each PHAB contained four separate tissue chambers, and the complete package of tissue chambers and PHAB created a compact platform that not only allowed for automated data collection and maintenance of proper cell culture conditions but was also easily manipulated for fluid exchanges and helped reduce crew workload (Fig. 2*F*).

Spaceflight-Induced EHT Contractile Dysfunction. In vivo, contractile forces of the heart must be sufficient to overcome arterial resistance during systole for successful delivery of blood throughout the body, and thus decreased cardiomyocyte twitch forces can be indicative of pathological cellular changes (24). After 12 d of spaceflight, EHTs exhibited a significant decrease in twitch force production relative to both baseline values and ground control tissues, and this decrease in twitch force was maintained throughout the duration of the experiment as seen by the negative trend in Fig. 3*A*. This decreasing trend was also reflected in the twitch force fold change of each individual spaceflight tissue as compared to its baseline recording, whereas average fold change values for ground control tissues were relatively consistent (Fig. 3*B*). However, the twitch force fold change differences were not always statistically significant between the two groups. This observed deficit in contractile function persisted after the 9-d period upon return to Earth.

To interpret the effect of spaceflight on cardiac function more deeply, we visualized the overlap of the baseline measurements compared to Day 39 as Gaussian distributions of the populations and reported the difference in the means of the populations using the statistical metrics Cohen's d, Cohen's U_3, and percent overlap

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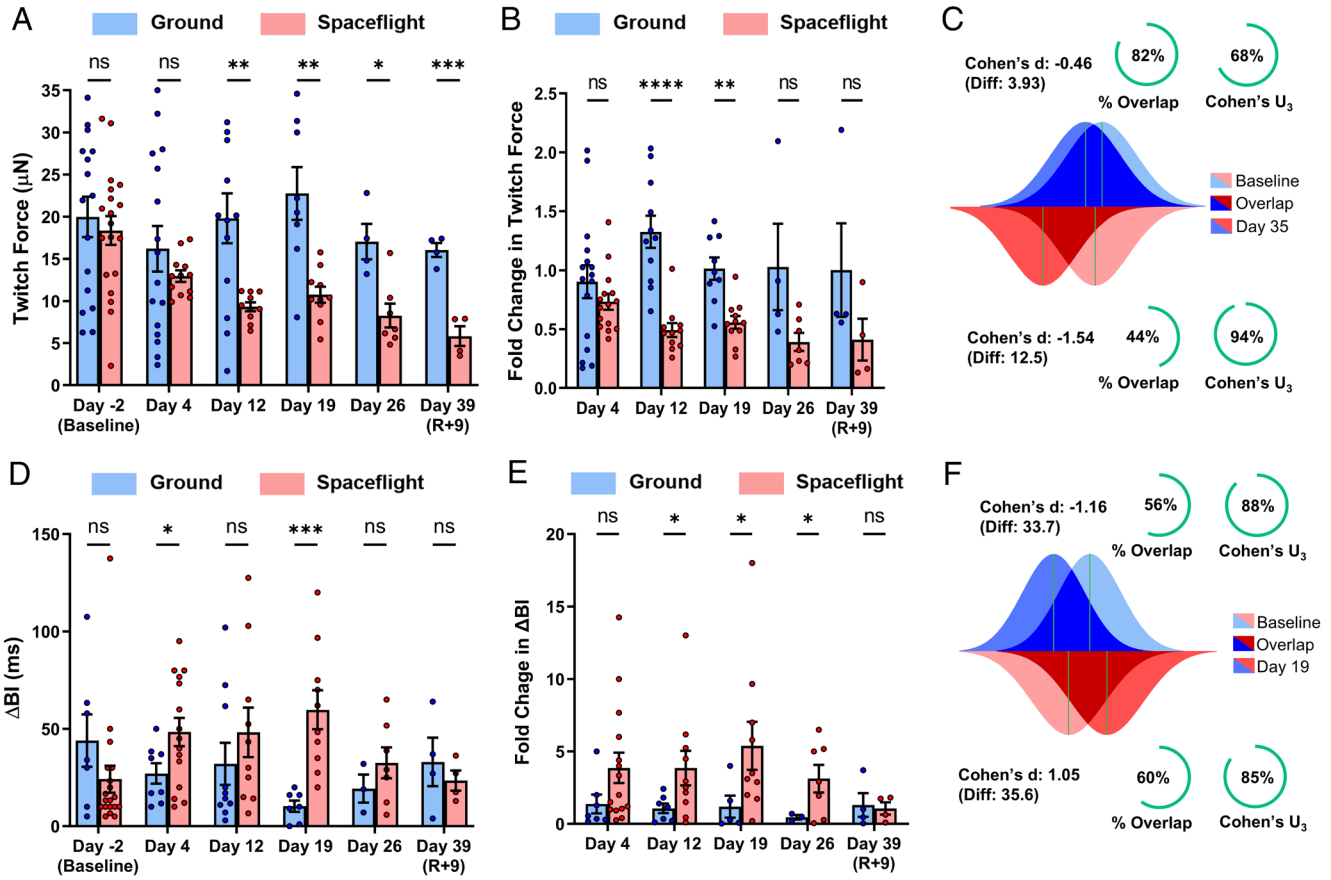


Fig. 3. Impaired force production and increased arrhythmia in spaceflight EHTs. (*A*) Raw twitch force and (*B*) fold change in twitch force over time for ground controls and spaceflight samples. (*C*) The fit Gaussian distributions for Day 39 and Baseline twitch force timepoints indicating the Cohen's d and difference in means, overlapping coefficient, and Cohen's U3 for spaceflight (red) and ground controls (blue). Green lines in the Gaussian distributions indicate the means of each dataset. (*D*) Fold change in the change in beat interval and (*E*) raw change in beat interval over time for ground controls and spaceflight samples. There were initially 16 tissues for ground controls and 20 for spaceflight samples. Over time, the numbers decreased due to the scheduled preservation of tissues for imaging or RNA-seq at each time point, resulting in n = 4 at the end of the experiment for both the ground control and spaceflight samples. (*F*) The fit Gaussian distributions for Day 19 and Baseline change in beat interval timepoints indicating the Cohen's d and difference in means, overlapping coefficient, and Cohen's U3 for spaceflight (red) and ground controls (blue). Green lines indicate the means of each dataset. Data are mean ± SEM. (*P < 0.05, **P < 0.01, ***P < 0.001, ****P < 0.0001, ns = no significance as determined by a Welch's t test) Time points chosen for further analyses (*C* and *F*) were chosen based on severity, as compensatory mechanisms identified later in the text may lead to recovery of interval irregularity at later timepoints.

for both spaceflight and ground control samples (Fig. 3*C*). Cohen's d indicates the magnitude of the difference between two groups and is canonically interpreted such that the perturbation has a small effect if the absolute value is 0.2 or less, a medium effect if between 0.2 and 0.5, and a large effect if 0.8 or larger, where negative values indicate a decrease in the means between the populations and a positive value indicates an increase (25). This analysis indicated that the ground controls had a medium-sized decrease in twitch forces at Day 39 as compared to baseline (Cohen's d = −0.46), which correlated with a decrease of 3.93 μN in the mean twitch force between Day 39 and baseline. Additionally, 82% of the distributions of Day 39 and the baseline overlapped, and 68% of the twitch force values on Day 39 were below the mean of the baseline measurements (Cohen's U_3). In contrast, the spaceflight samples exhibited a much larger average decrease in twitch force of 12.5 μN at Day 39 as compared to baseline (Cohen's d = −1.54), only 44% of the distributions of Day 39 and baseline overlapped, and 94% of the twitch force values at Day 39 were below the mean of the baseline measurements. The decreasing trend found in the fold change in twitch forces (Fig. 3*B*) supported the rationale that this decrease is caused by spaceflight-induced damage.

Given that arrhythmias were the second-most frequent medical problem for Russian astronauts during the Mir program (26), we sought to determine whether similar dysfunction was present in the spaceflight EHTs. To do so, the mean change in beat interval (ΔBI) was quantified, where an increased ΔBI indicates greater beating frequency irregularity. Indeed, the ΔBI was found to be elevated in spaceflight samples compared to controls while these tissues were still on board the ISS; however, this arrhythmic behavior appeared to resolve after return to Earth (Fig. 3*D*). Similarly, individual tissue fold changes in ΔBI indicated that there were significant increases in ΔBI in space-flown EHTs compared to ground controls for the duration of the spaceflight mission, and these values returned to normal upon return to Earth gravity (Fig. 3*E*). For ΔBI analysis, the Cohen's d and other metrics were calculated for Day 19 and the baseline recordings, instead of Day 39 as before, as this was the latest time point at which a statistically significant increased ΔBI was present in the spaceflight tissues. Reduced arrhythmic activity at the later stages of spaceflight could have been due to the activation of compensatory mechanisms for spaceflight-induced damage that are described later in this study. Ground controls had a very large decrease in ΔBI of 33.7 ms at Day 19 compared to baseline (Cohen's d = −1.16), 56% of the distributions of Day 19 and baseline overlapped, and 88% of the ΔBI values at Day 39 were below the mean of the baseline measurements (Cohen's U_3) (Fig. 3*F*). These values indicated that as the ground control EHTs matured, they had a substantially lower

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arrhythmic potential. In contrast, the spaceflight samples had a very large increase in ΔBI of 35.6 ms on Day 19 as compared to baseline (Cohen's d = 1.05), 60% of the distributions of Day 19 and baseline overlapped, and 85% of the ΔBI values at Day 19 were higher than the mean of the baseline measurements (Cohen's U_3) (Fig. 3*F*). This striking difference in comparison to the ground controls indicated a substantial increase in arrhythmic potential in spaceflight samples.

Spaceflight-Induced Disruption of Cardiomyocyte Contractile Apparatus. Sarcomeres are the basic contractile unit within muscle cells, and the maximal twitch forces of a cardiomyocyte are governed in part by the length of the sarcomere (24). Decreased cardiac output and function during heart failure have also been linked to sarcomere dysfunction (27). Ultrastructural images of spaceflight and ground control sarcomeres were taken using transmission electron microscopy (TEM) and analyzed to determine whether the observed contractile dysfunction induced by spaceflight may be linked to changes in sarcomere structure (Fig. 4*A*). It was found that sarcomere length and Z-band width were both significantly decreased in spaceflight samples that were fixed on Day 39 (Fig. 4 *B* and *C*), indicating that spaceflight had induced deleterious structural modifications to sarcomeres, which did not recover upon return to normal gravity, possibly explaining the noted inability of these tissues to regain twitch force production after return to Earth.

Ultrastructural Imaging Reveals Disruptions to Mitochondrial Morphology in Spaceflight EHTs. Changes in mitochondrial structure and indications of mitochondrial dysfunction were also evident in ultrastructural TEM images. Mitochondria in healthy cardiomyocytes are typically in large networks with developed cristae; these morphological features are required for the high oxidative capacity of energetically demanding heart

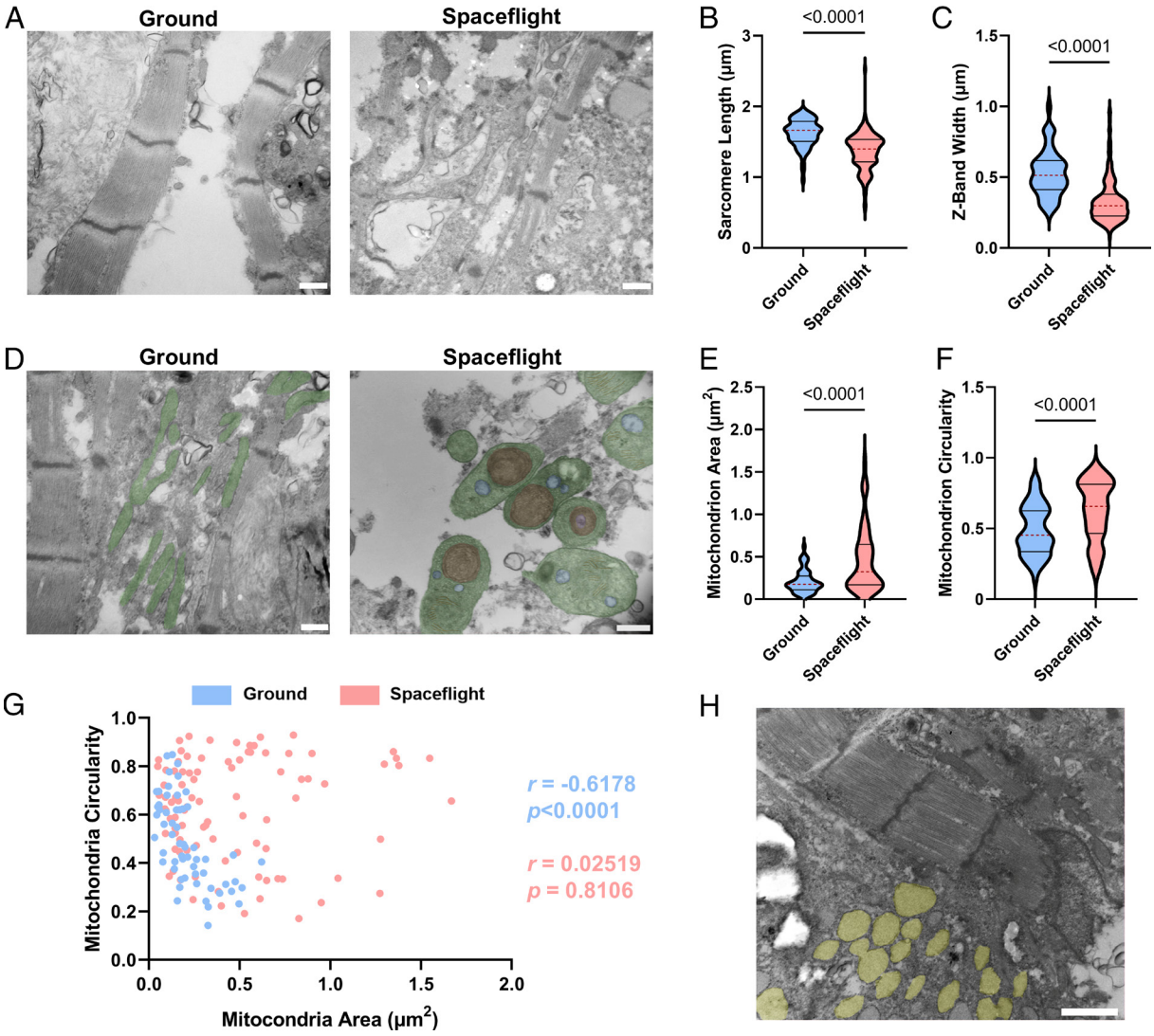


Fig. 4. Ultrastructural images and analysis reveals perturbations to sarcomere structure and mitochondrial function. (*A*) Representative TEM images of sarcomeres from hiPSC-CMs in Day 39 EHTs from ground and spaceflight experiments. (Scale bar, 500 nm.) (*B*) Analysis of sarcomere length and (*C*) Z-band width from for Day 39 spaceflight and control samples. The red dashed line indicates the median and the black solid lines indicate the upper and lower quartiles. n = 67 sarcomeres for ground controls (4 tissues, 15 to 17 replicates each) and n = 144 sarcomeres for spaceflight (4 tissues, 36 replicates each), *P*-values shown determined with Welch's *t* test. (*D*) Representative TEM images of mitochondria (green) for Day 39 spaceflight and control samples. Red indicates circular cristae, and blue indicates compartmentalization in the cristae. (Scale bar, 500 nm.) (*E*) Analysis of mitochondrion area and (*F*) mitochondrion circularity. The red dashed line indicates the median; the black solid lines indicate the upper and lower quartiles. (*G*) Plot of mitochondrion circularity against area, with the Pearson Correlation Coefficient (r) and associated *P*-value shown. n = 58 mitochondria for ground controls (4 tissues, 15 to 17 replicates each) and n = 93 mitochondria for spaceflight (4 tissues, 36 replicates each), *P*-values shown in *E* and *F* determined with Welch's *t* test. (*H*) Representative TEM image of intracellular lipid accumulations (yellow) found in Day 39 spaceflight samples. (Scale bar, 500 nm.)

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tissue (28). However, mitochondria in spaceflight samples were noticeably fragmented, and mitochondrial swelling was apparent (Fig. 4*D*). This excessive swelling often indicates an opening of the mitochondrial permeability transition pore and depends on high levels of intracellular Ca^{2+} and reactive oxygen species (ROS) (29). ROS are chemically reactive molecules that contain oxygen and are formed as by-products of normal cellular metabolism. ROS can cause damage to cellular components such as DNA, proteins, and lipids, leading to oxidative stress and inflammation. Mitochondrial swelling also plays a key role in mitochondrial-mediated cell death pathways (30), and pathological mitochondrial swelling is associated with oxidative stress and is found in cardiac ischemia–reperfusion injuries (31). Furthermore, fragmentation and swelling of mitochondria are both associated with aging (32), diabetes (33), and inflammation (34). In general, mitochondrial dysfunction is closely tied to heart failure and cardiac arrhythmias (35). To quantitatively assess mitochondrial swelling and fragmentation, both area and circularity of individual mitochondria were measured in spaceflight and ground control samples. Circularity was defined such that a value of 1 indicates a perfect circle and a value of 0 indicates a straight line. Mitochondria were both significantly larger (Fig. 4*E*) and more circular (Fig. 4*F*) in spaceflight samples. By plotting circularity vs. area of the mitochondria and fitting a simple linear regression to the data, it was revealed that ground control samples had a statistically significant ($P < 0.0001$) and negative (Pearson's correlation coefficient of −0.62) correlation between circularity and area (Fig. 4*G*). This indicated that larger mitochondria in ground control tissues tended to be less circular and were therefore likely to be networks of healthy mitochondria. However, this same analysis indicated that no such correlation existed in the spaceflight samples, where many of the larger mitochondria were fragmented, swollen, and altogether morphologically abnormal.

Additional signs of mitochondrial ultrastructural defects were found in spaceflight tissues. Defects in mitochondrial ultrastructure have been cataloged previously and shown to be involved in mitochondrial-related myopathies (36). Abnormalities reported include paracrystalline inclusions, cristae linearization, concentric cristae, and compartmentalization. When analyzing the mitochondria, many of these same ultrastructural defects were found in the mitochondria. Indeed, over 50% of the mitochondria analyzed in spaceflight samples, but fewer than 10% of mitochondria in control samples, exhibited ultrastructural abnormalities (images available in the Microphysiology Systems Database). Numerous instances of excessive intracellular lipid droplet accumulation, a hallmark of mitochondrial dysfunction (37), were also identified in spaceflight samples (Fig. 4*H*). Taken together, these images and analyses indicate high instances of mitochondrial fragmentation, swelling, and abnormal structural changes due to spaceflight. In fact, mitochondrial dysfunction has recently gained attention as a potential central hub for spaceflight-induced damage across multiple species and tissue types, though the cause of this damage remains unclear (38).

RNA Sequencing (RNA-Seq) Canonical Pathway Analysis Suggests Spaceflight Induces Perturbations of Pathways Critical for Normal Cardiac Function. To gain further insight into the mechanism of cardiac dysfunction of EHT spaceflight samples, RNA-seq was performed on ground control and spaceflight samples from Day 26 and Day 39. A principal component analysis indicated high levels of separation between ground and spaceflight clusters, indicating clear differences in the populations (*SI Appendix*, Fig. S1). The most differentially expressed genes were identified (Fig. 5*A*) and these genes were loaded into Ingenuity Pathway Analysis (IPA) software to generate an interaction map of the canonical pathways that appeared to be most significantly modified due to spaceflight, where nodes colored blue were down-regulated or inhibited, nodes colored orange were up-regulated or activated, a solid line represents a direct interaction, and a dotted line represents an indirect interaction (Fig. 5*B*). Additionally, a dot plot was generated to indicate z-score and the –log(*P*-value) to highlight the significance and degree of modulation of several of these identified pathways (Fig. 5*C*).

Down-regulation of pathways associated with calcium signaling, cardiac hypertrophy signaling, formation of muscle, and cardiogenesis were supported by the analyses and are consistent with the impaired contractile function and the observed presence of significant defects in the cardiac contractile apparatus of spaceflight EHTs. Changes in calcium signaling due to spaceflight were also found in a previous study with cardiomyocyte monolayers (14) and the down-regulated pathways are consistent with the observed arrhythmias and changes in mitochondrial function and structure in this study. Down-regulation of fatty acid metabolism and PPAR signaling is indicative of mitochondrial dysfunction and impaired lipid oxidation, which could explain the observed lipid droplet accumulation. Together these deficits could lead to overall cardiac dysfunction through the inability of cardiomyocytes to generate enough ATP to meet cellular demands.

Up-regulated pathways include those for glucose metabolism disorders, dilated cardiomyopathy, and heart failure. Up-regulation of genes associated with glucose metabolism disorder further illustrated mitochondrial dysfunction resulting from spaceflight. One of the hallmarks of dilated cardiomyopathy is impaired cardiac contractility (39), and this also agrees with the functional data collected from the spaceflight tissues. Overall, these differentially expressed and regulated pathways support the functional and ultrastructural findings suggesting that spaceflight and prolonged exposure to microgravity induces significant negative changes in cardiac physiology, potentially through oxidative stress and changes in calcium handling, with the net result being progressive heart failure.

Interestingly, numerous inflammatory pathways (INFG, TNF, IL1B, IL18, IFNA2, IFNL1) were overrepresented due to the up-regulation of pattern recognition pathways, including cGAS and MAVS. Cyclic GMP-AMP synthase, or cGAS, is well known to recognize mitochondrial DNA (mtDNA) released from functionally compromised mitochondria and initiates an immune response through formation of cGAMP and activation of STING (40–42). cGAS is also known to have major impacts on lipid metabolism and mitochondrial health (43). Mitochondrial antiviral-signaling protein (MAVS) is located in the outer membrane of a mitochondrion and, upon activation by ROS during oxidative stress (44), forms aggregates on the mitochondrion membrane and activates a signaling cascade resulting in inflammatory cytokine up-regulation (45). These findings indicate that mitochondrial dysfunction in spaceflight samples may also lead to downstream inflammatory responses that further exacerbate the deleterious effects of spaceflight on cardiac function and health.

RNA-Seq Results Indicate That Spaceflight Perturbs Genes Associated with Contraction, Oxidative Stress, and Inflammation. Specific genes related to cardiac contraction, calcium handling, mitochondrial function, and inflammation were further analyzed in spaceflight and ground control EHTs. Genes encoding proteins required for contractile activity were significantly down-regulated in spaceflight, including MYL7, MYL3, MYH7B, TNNI3, and TTN (Fig. 5*D*). MYL7, MYL3, and MYH7B all encode myosin isoforms involved in cardiac muscle contraction. TNNI3 encodes

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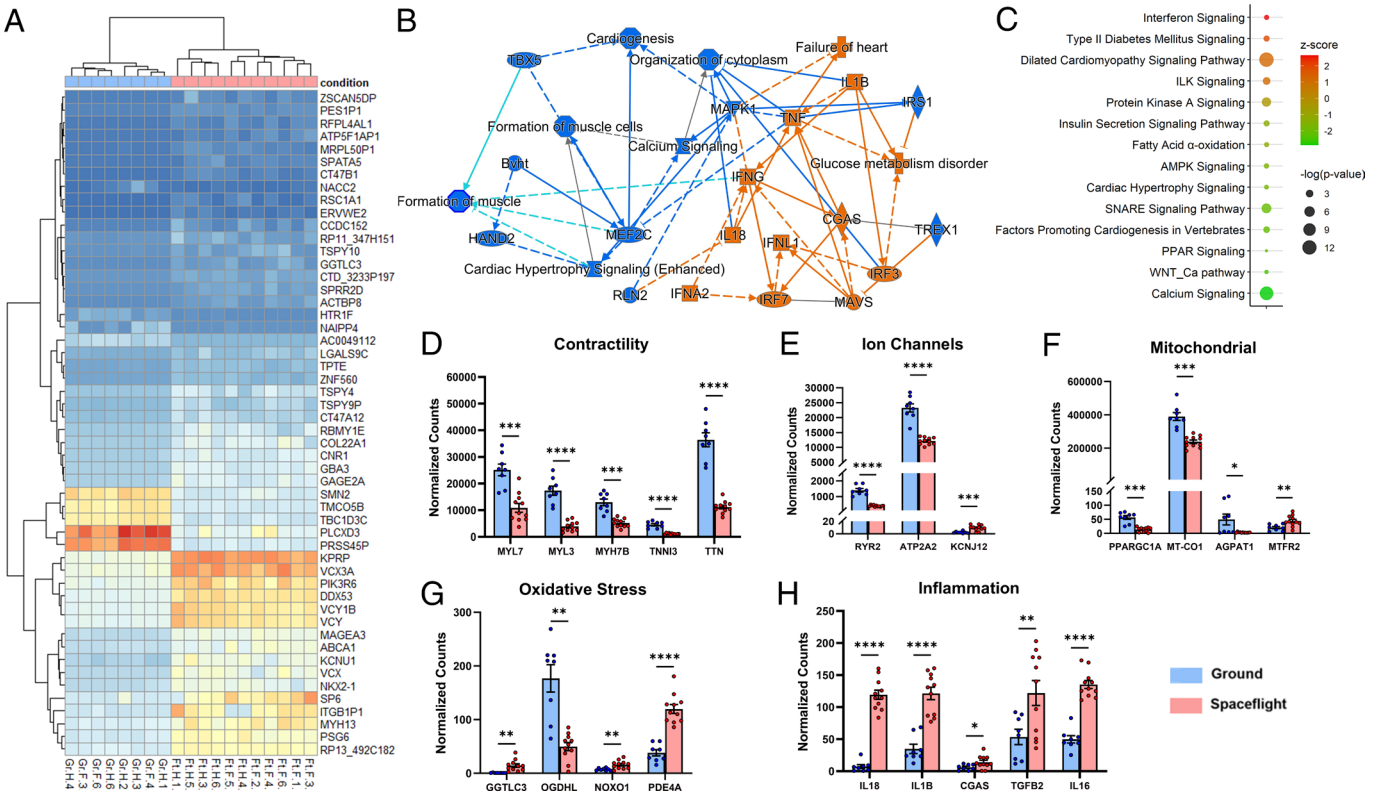


Fig. 5. RNA-seq analysis of spaceflight and ground control EHTs reveals significant expression level diversion of key genes and pathways. (*A*) Heatmap of the top differentially expressed genes with log2-fold change absolute values >4. (*B*) Canonical pathways projected to be of highest importance through IPA. Nodes colored blue were down-regulated or inhibited, nodes colored orange were up-regulated or activated, a solid line represents a direct interaction, and a dotted line represents an indirect interaction. Light blue lines indicate less confidence and dark blue lines indicate higher confidence. (*C*) Some of the top canonical pathways identified using IPA are visualized by activation z-scores (dot color) and –log(*P*-values) (dot size). Positive z-score indicates that the pathway is activated, and negative z-score indicates that the pathway is inhibited. (*D*) Normalized counts for genes associated with contractility, (*E*) ion channels, (*F*) mitochondrial function and metabolism (*G*) oxidative stress, and (*H*) inflammation. n = 11 tissues for spaceflight and n = 9 tissues for ground controls. (*$P < 0.05$, **$P < 0.01$, ***$P < 0.001$, ****$P < 0.0001$ as determined by a Welch's *t* test.)

cardiac troponin I, which is a subunit in the troponin–tropomyosin complex that regulates the contractile complex in cardiac muscle, and is depleted in ischemic (46) and infarcted myocardium (47). TTN encodes titin, which is responsible for the passive elasticity of relaxed muscle and, therefore, for the restoring force during diastole. The down-regulation of these genes likely contributed to both the contractile dysfunction and the significant changes in sarcomere structure found in spaceflight EHTs.

Likewise, the RYR2 and ATP2A2 genes encoding ion channels critical for proper cardiac excitation–contraction coupling were both significantly down-regulated in spaceflight (Fig. 5*E*). Ryanodine receptor 2 (encoded by RYR2) is the major mediator for sarcoplasmic release of stored calcium ions into the cytosol during calcium-induced calcium release, while SERCA (encoded by ATP2A2) is responsible for transfer of cytosolic calcium ions back into the sarcoplasmic reticulum during muscle relaxation. Down-regulation of these critical regulators of calcium homeostasis can lead to cardiac arrhythmias like those observed in the spaceflight EHTs (48). In contrast, KCNJ12 was found to be up-regulated in spaceflight samples. KCNJ12 encodes an inwardly rectifying potassium channel that is commonly expressed in cardiomyocytes and is responsible for regulating the depolarization and repolarization of the membrane potential. Up-regulation of KCNJ12 appears in patients with chronic atrial fibrillation (49), so it may have also contributed to the arrhythmic behavior of the spaceflight EHTs.

Three genes associated with metabolism and mitochondrial function were significantly down-regulated in spaceflight samples (Fig. 5*F*). PPARGC1A encodes PPARG coactivator 1α (PGC1-α), a key regulator of energy metabolism, stimulator of mitochondrial biogenesis, and promoter of glycolytic/oxidative metabolism during normal cardiac muscle development (50). Furthermore, AGPAT1 was also significantly down-regulated which has been shown to decrease glycogen synthesis in muscle cells (51). These two shifts indicate a potential deficiency in both oxidative phosphorylation and glucose metabolism in spaceflight EHTs. MT-CO1 encodes cytochrome-c oxidase I (COX1), the main subunit of the cytochrome-c oxidase complex, which is the last enzyme in the electron transport chain for the synthesis of ATP. Any deficiencies in ATP stores caused by down-regulation of this gene would thereby contribute to overall cellular dysfunction. Conversely, it was found that mitochondrial fission regulator 2 (MTFR2) was up-regulated in spaceflight tissues (Fig. 5*F*). MTFR2 promotes mitochondrial fission (52), and therefore may have contributed to the mitochondrial fragmentation found in spaceflight samples. Overall, these results indicate that spaceflight may result in deleterious modifications in metabolic function, ATP production, and mitochondrial morphology.

Genes associated with oxidative stress were also closely examined as increases in ROS may have caused numerous downstream negative effects on cellular function (Fig. 5*G*). Among the most up-regulated genes was GGTLC3, which encodes gamma-glutamyltransferase-1 (GGT1), an enzyme that initiates cleavage of reduced and oxidized glutathione into glutamate, cysteine, and glycine (53). Cysteine availability is the rate-limiting step in the synthesis of glutathione (54), an important antioxidant that mitigates ROS-induced oxidative damage. Up-regulation of GGTCL3 at these later timepoints may indicate that oxidative stress was caused by spaceflight and cells then

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increased GGT1 synthesis to compensate for this damage. This could also have led to the observed decrease in ΔBI upon return to Earth gravity, as oxidative stress is known to contribute to arrhythmias (55) and ROS directly interacts with and modifies the action of numerous ion transport channels (56). NOXO1 was also up-regulated in spaceflight tissues and encodes NADPH oxidase organizer 1, which positively regulates NOX1 and NOX3 and leads to increased generation of ROS and potential oxidative stress (57). cAMP-specific 3′,5′-cyclic phosphodiesterase 4A (PDE4A) encodes an enzyme by the same name that is involved in cAMP signaling, and inhibition of both PDE4A and cAMP has been shown to help mitigate oxidative stress, while overexpression increases oxidation (58, 59). The significant up-regulation of these genes indicates oxidative stress may have resulted from spaceflight and that perhaps, most interestingly, the cardiomyocytes within the EHTs may have compensated for this over time.

Finally, it was found that the inflammation-associated genes IL-18, IL-1β, cGAS, TGFβ-2, and IL-16 were all significantly up-regulated in spaceflight samples (Fig. 5*H*). Interleukin 18 (IL-18) (60), IL-1β (61), and IL-16 (62) are inflammatory cytokines that serve key roles in pathogenesis following myocardial infarction, and IL-18 in particular serves as a clinical predictor of cardiovascular death in patients with coronary artery disease (63). IL-1β additionally induces mtDNA release to activate cGAS signaling (64) and is up-regulated by MAVS (65). TGF-β2 is known to be activated by increased ROS levels (66), exacerbates diabetic cardiomyopathy through pathophysiological remodeling (67), regulates mitochondrial oxidative stress response (68), and may contribute to mitochondrial fission and fragmentation (69). These data together indicate that spaceflight results in an up-regulation in inflammatory signaling, potentially through mitochondrial and oxidative stress-related pathways.

In Silico Simulations Reveal ROS Activation Potentially Leads to the Canonical Pathway Modifications Caused by Spaceflight. Mitochondrial dysfunction has been implicated as a primary effect of spaceflight across numerous tissue models and as a mediator of downstream spaceflight-induced tissue damage (38). Oxidative stress occurs with, and contributes to, mitochondrial dysfunction, which can lead to cardiovascular dysfunction and inflammation. To assess the hypothesis that spaceflight-induced oxidative stress and mitochondrial dysfunction may contribute to these deleterious changes in space flown EHTs, we utilized the Molecular Activity Predictor in silico modeling capabilities of the IPA pipeline (*SI Appendix*, Fig. S2). Here, ROS presence was artificially increased, and downstream pathways were connected. Predicted activation and inhibition of these downstream pathways occurred based on the increased presence of ROS and pathway interconnectivity. Our RNA-seq analysis measurements were then overlaid on these predictions to determine whether the predicted activation or inhibition of pathways in the Molecular Activity Predictor was the same as actual activation or inhibition in the experimental results, and all findings were consistent across the model and experimental results, indicating that increased ROS presence and oxidative stress can lead to modifications in transcriptional regulation as seen in spaceflight.

Discussion

Astronauts have been found to have decreased ventricular volume, decreased cardiac output, and cardiac muscle atrophy after spaceflight (3). In this study with EHTs, spaceflight was found to cause impaired contractility, detrimental subcellular structural modifications, mitochondrial structural changes, and increased oxidative stress. RNA-seq analysis of spaceflight samples indicated transcriptional changes associated with metabolic dysfunction, increased inflammatory cytokine production, and heart failure pathway up-regulation. Additional RNA-seq analysis and in silico modeling indicated that oxidative stress and mitochondrial dysfunction may have led to downstream tissue damage and cardiovascular dysfunction.

In the context of astronauts, exposure to space radiation and microgravity can increase ROS production and exacerbate oxidative stress (6, 70), which may contribute to a range of health issues including cardiovascular disease, neurodegeneration, and cancer. Oxidative stress can both contribute to and be caused by mitochondrial dysfunction (71). ROS can additionally damage proteins, lipids, and both DNA and RNA, leading to ROS-induced ROS release, initiating a vicious cycle. ROS can also modify cardiomyocyte contractility through interaction with ion channels or pumps, (72) and mitochondrial dysfunction is well known to contribute to cardiomyopathies (35). Both of these can initiate inflammatory activation (40, 44). While this is supported by the transcriptomic, functional, and structural data presented in this study, additional work will be needed to confirm the presence of oxidative stress and mitochondrial dysfunction in space-flown EHTs and to evaluate preventative or therapeutic strategies for this spaceflight-induced pathology.

It has been previously found that tissue complexity modifies the degree of cellular damage resulting from spaceflight, indicating that monolayer-based studies are insufficient for replicating the full spectrum of spaceflight pathophysiology (38). Therefore, the commonality of the findings from this study with prior observations in humans (6, 73, 74) illustrates the capability of our enhanced maturity EHTs to accurately model many of the detrimental effects of spaceflight on human cardiac function in a relatively high-throughput manner. Despite these promising results, our model cannot yet factor in the aforementioned changes in mechanical loading and changes in arterial blood pressure (75, 76) that arise from microgravity. Indeed, one weakness of this and other studies of spaceflight effects on in vitro muscle and cardiovascular systems is that the resistance to contraction remains essentially the same in space as on Earth in these systems. Coupling the methods used in this study (to improve the maturation of hiPSC-CMs) with advancements in tissue engineering techniques, such as 3D bioprinting, that allow for the fabrication of heart chamber models (77), could seek to address this in future studies. It should also be noted that this study's observed effects on cardiac function and transcriptomic changes are likely due to both microgravity and radiation. While it is difficult to ascertain what role each factor played in causing the various forms of dysfunction in this study, future missions with modified platform designs or ground-based experiments with controlled radiation exposures can begin to answer this question effectively.

The similarities between aging and spaceflight have been noted in previous studies (78, 79). Both aging and spaceflight cause a decline in cardiovascular function, with changes in heart structure and function leading to an increased risk of cardiovascular disease. Telomere shortening was observed as a result of spaceflight in the NASA Twins Study and is known to occur in aging (6). ROS is a well-known contributor to the aging process, causing DNA damage and metabolic dysfunction (80). In fact, the indications of mitochondrial dysfunction shown in this study are the same as those that occur in aging (81), namely decreases in mitochondrial biogenesis, mitochondrial swelling, and abnormalities in mitochondrial quality control, with oxidative stress as a contributing factor. The presence of these exacerbating factors demonstrates

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the value of spaceflight as a research environment for not only studying astronaut health, but for also studying aging in an accelerated timeframe.

Methods

Preparation of dECM-rGO Composite Hydrogel Scaffolds. Electroconductive and bioactive composite hydrogel scaffolds were prepared as previously described (19). The left ventricular myocardium was isolated from freshly slaughtered adult pig hearts and processed into small slices approximately 1 mm in thickness. Slices were placed in a solution of 1% w/v sodium dodecyl sulfate (SDS; Fisher Scientific) for 36 h, with an additional wash in 1% w/v Triton-X100 (Sigma-Aldrich) for 1 h to complete cell removal. This was followed by washes in phosphate-buffered saline (PBS) over 3 d, with the PBS changed every day to remove excess detergent and cell debris. The resulting decellularized extracellular matrix (dECM) was sterilized with 0.1% w/v peracetic acid in 4% w/v ethanol, followed by an additional overnight wash in PBS. dECM was then lyophilized and stored at 4 °C until used.

rGO was chemically reduced from commercially available graphene oxide solution (GO; Graphene Laboratories) using $NaBH_4$ (Sigma-Aldrich). GO flakes with a lateral flake size of 90 to 200 nm in a 1 mg/mL stock solution were mixed with 300 mM $NaBH_4$ for 1 h at room temperature. The rGO was then vacuum filtered through 0.1 μm pore size mixed cellulose ester membranes (MilliporeSigma) and resuspended in dH_2O at 10 mg/mL with bath sonication.

Lyophilized dECM was solubilized by mixing with pepsin at a 10:1 mass ratio in 0.1 M HCl; rGO was also added at this stage. A 2% w/v dECM with 0.3% w/v rGO mixture was used in this study. Mixtures were stirred until the dECM-rGO was completely solubilized and had formed a homogenous solution, at which point the pH was brought up to 7.4 with 10 M NaOH. 10X RPMI 1640 (Sigma-Aldrich) was added at 10% of the final volume to restore isotonic balance. Hydrogel solutions were stored at 4 °C until use.

hiPSC Maintenance and Differentiation. UC 3-4 hiPSCs (deidentified and produced with approval from an IRB, generous gift from David Mack, Department of Rehabilitation Medicine, University of Washington, Seattle WA 98104) (82) were cultured and differentiated using small molecule modulation of Wnt/β-catenin signaling (83). hiPSCs were expanded and maintained on Matrigel-coated plates in mTeSR1 medium (STEMCELL Technologies) prior to induction. Induction was achieved by placing the cells in RPMI 1640 medium (Invitrogen) supplemented with B27 without insulin (Invitrogen) and 10 μM CHIR-99021 (Selleck Chemicals) for 18 h, after which the medium was replaced with fresh RPMI/B27 without insulin for 48 h. The medium was then replaced with RPMI/B27 without insulin and 5 μM IWP-4 (REPROCELL), and cells were cultured for 48 h, after which the medium was once again replaced with fresh RPMI/B27 without insulin for 48 h. Finally, the medium was replaced with RPMI/B-27 (Invitrogen), and cultures were fed with fresh medium every other day. Spontaneously beating cells were typically observed 10 to 12 d after induction. Differentiation runs that produced a hiPSC-CM population of >90% positive for human cardiac troponin T (cTnT) as measured by flow cytometry were used for all experiments.

Human Stromal Cell Culture and Maintenance. HS-27A human bone marrow-derived stromal cells (deidentified and acquired through procedures approved by an IRB (84), purchased from ATCC) were cultured and maintained in high-glucose DMEM (Invitrogen) supplemented with 10% fetal bovine serum.

Engineered Heart Tissue Lattice Fabrication. EHT post arrays for the heart-on-a-chip platform were fabricated using techniques described previously (23). Polydimethylsiloxane (PDMS; Sylgard 184, Dow Chemical) at a 10:1 base to curing agent ratio was poured into an acrylic mold. A glass capillary tube was inserted into one post of each pair to impart rigidity to that post, and a 1 mm^3 neodymium magnet was placed in the opposing post cap. Molds were allowed to degas prior to being placed into a 65 °C oven overnight to cure the PDMS. Cured post arrays were then removed from the mold. Arrays were fabricated in sets of six pairs, with each post 12.5 mm long and 1.5 mm in diameter and featuring a cap structure 0.5 mm thick and 2 mm in diameter. Post-to-post spacing was 8 mm. Wells for tissue casting were fabricated by pouring 1 mL of PDMS into the wells of a 24-well culture plate before inserting custom 3D printed molds that were 12 mm × 4 mm × 4 mm (length × width × depth). After curing the PDMS overnight, molds were removed. All fabricated hardware was sterilized with 70% v/v ethanol and UV light prior to use.

Spaceflight Hardware. The heart-on-a-chip flight chamber is a fully contained enclosure for spaceflight, with each housing EHT arrays of 6 tissues. The chamber was composed of a polysulfone frame that featured a gas-permeable PDMS membrane on one side and a polyetherimide (Ultem®) cover on the other that was screwed tightly into place in combination with a rubber O-ring to prevent fluid leakage. Both ends of the chamber had self-sealing septa that served as syringe needle access ports for fluid exchange. Fluid exchanges were accomplished by attaching a syringe to each port and then filling the chamber with new culture medium or preservative while simultaneously removing old medium. Each chamber had a maximum volume of 30 mL.

The PHAB is an aluminum box that can be sealed or vented. Each PHAB housed 4 individual EHT culture chambers along with giant magnetoresistance (GMR) sensors and electronics mounted on a printed circuit board beneath the chambers to facilitate data collection and management. When transporting live EHTs to the ISS, the PHAB was charged with 5% CO_2 via overnight storage in an incubator with the ports left open. Ports and vents were then sealed prior to transport. Once sealed, the PHAB can maintain 4.5 to 5% CO_2 for up to 9 d. Once on board the ISS, the vent cover was removed and the PHAB was placed in the Space Automated Bioproduct Laboratory (SABL) on-board incubator that maintained 37 °C and 5% CO_2 culture conditions and provided power to the PHAB's electronics. Temperature set points for SABL were controlled from the ground via BioServe's Payload Operations and Command Center.

Formation and Maintenance of Engineered Heart Tissues. hiPSC-CMs were thawed and recovered on Matrigel-coated plates for 7 d prior to being casted into EHTs. hiPSC-CMs and HS-27A stromal cells were mixed in dECM-rGO gel solutions at cell densities of 20×10^6 cells/mL and 5×10^6 cells/mL, respectively. 70 μL of this cell-hydrogel mixture was then pipetted into each PDMS well, after which PDMS post arrays were placed into the wells. The 24-well plate was then placed into an incubator at 37 °C and 5% CO_2 for approximately 60 min, after which the hydrogels had crosslinked and 1 mL of RPMI/B-27 supplemented with Y-27632 dihydrochloride ROCK inhibitor and 1% penicillin/streptomycin was added to each well. After 24 h, the media was replaced with fresh RPMI/B-27. Subsequent media changes were conducted every other day.

Spaceflight and Ground Control Experiments. One week after casting, post arrays with formed tissues were transferred into individual heart-on-a-chip flight chambers. Tissues were hand-transported to the Space Station Processing Facility labs at Kennedy Space Center in a 37 °C and 5% CO_2 environment, where they were then cultured and allowed to mature for an additional 14 d, with media exchanges occurring every other day. Heart-on-a-chip flight chambers were loaded into PHABs charged with 5% CO_2 and handed over for capsule loading 2 d prior to launch as part of the SpaceX CRS-20 resupply mission. Mission payload, including prefilled 30 mL syringes containing RPMI/B-27 stored at 4 °C, arrived at the ISS 2 d after launch, and the first media exchange was performed 2 d later (4 d postlaunch). Contractile activity of EHTs was continuously monitored and recorded in real-time using the postembedded neodymium magnets and GMRs as described previously (23), with media exchanges and preservation of EHT samples in either 4% paraformaldehyde (PFA) in PBS or RNALater (Thermo Fisher Scientific) conducted at 12-, 19-, and 26-d postlaunch. Samples fixed with PFA or preserved in RNALater were stored at 4 °C or −80 °C, respectively, for the entire duration of the study. After 30 d in low Earth orbit, the remaining live tissues and preserved samples were returned to Earth. Two days after capsule splashdown, live EHTs were retrieved and hand-transported to the lab at the University of Washington in a 37 °C and 5% CO_2 environment and were cultured for an additional 7 d in a 1G environment before preservation (39 d postlaunch; 9 d post-Earth return). Ground control experiments were fully conducted at 1G on an identical culture and media exchange timeline, with all transportation steps and conditions replicated as closely as possible.

TEM of EHTs. Cryostat sections (10 to 20 μm) were air-dried for 30 min at room temperature prior to TEM sample preparation. Slides were fixed in ice-cold 2.5% glutaraldehyde, 3 mM $MgCl_2$, in 0.1 M sodium cacodylate buffer (pH 7.2) for 48 h at 4C. After buffer rinse, samples were postfixed in 1% osmium tetroxide in 0.1 M sodium cacodylate (for 1 h) on ice in the dark. Following a dH_2O rinse,

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slides were incubated in 2% uranyl acetate (UA) for 30 min, followed by dehydration in a graded series of ethanol, followed by propylene oxide and embedded in Eponate 12 (Ted Pella) resin. Samples were polymerized at 60C overnight on inverted BEEM capsules. Slides were separated from BEEM capsules by full immersion in liquid nitrogen.

Thin sections, 60 to 90 nm, were cut with a diamond knife on a Reichert-Jung Ultracut E ultramicrotome and picked up with 2 × 1 mm formvar coated copper slot grids. Grids were stained with 2% uranyl acetate in 50% methanol and 0.4% lead citrate before imaging on a Hitachi 7600 TEM at 80 kV. Images were captured with an AMT XR80 CCD (8 megapixel) camera. Morphological analysis of sarcomeres and mitochondria was performed using ImageJ Fiji software.

RNA Extraction and Sequencing Analysis. EHTs preserved in RNALater were gently thawed on ice until tissues could be transferred to bead homogenizer tubes and homogenized. Tissue lysates were then processed using the PARIS™ Kit (ThermoFisher Scientific) according to the manufacturer's protocol. The purified RNA libraries were prepared for standard RNA-seq and rRNA were removed using poly-A selection according to GeneWiz procedures. The sequencing data were preprocessed using Trim Galore (85) to remove the adapter regions and reads with mean base Phred scores below 30. The first 12 bases at the 5′ end and the first 10 bases at the 3′ end of each sequence read were clipped. The reads were aligned to the prebuilt Ensembl Transcriptome v96 using Kallisto (86) and differential expression between the EHTs from space flight and ground control groups was analyzed using DESeq2 (87). Differentially expressed genes (DEGs) with total unnormalized counts less than 100 or with normalized counts greater than or equal to five in less than three samples were filtered out. DEGs with at least twofold differences between the two groups and adjusted *P* value less than 0.05 were submitted to Ingenuity Pathway Analysis (IPA; Qiagen) to perform functional analysis.

In Silico Modeling and Analysis. IPA (Qiagen) was used to predict whether oxidative stress and mitochondrial dysfunction could lead to the downstream canonical pathway modifications observed. Briefly, the Molecule Activity Predictor function was utilized to connect a signaling cascade of related pathways with up- or down-regulation determined by our RNA-seq data. ROS presence was then added to the signaling cascade and artificially turned on. The Molecule Activity Predictor was then used to determine whether ROS activation alone was sufficient to lead to our RNA-seq results of up- or down-regulation of the pathways chosen. Any results that disagree with the RNA-seq data would be shown by a yellow line.

Statistical Analyses. Unless otherwise noted, all quantitative data are presented as means ± SE with a minimum of n = 6. One-way ANOVA with a Tukey's post hoc test was used to analyze datasets that included more than two experimental groups. Comparisons between two variables were performed using an unpaired *t* test with a Welch's correction, as SD could not be assumed to be equal. Cohen's d was calculated as $\delta = \frac{\mu_2 - \mu_1}{\sigma}$, where δ is the population size, μ are the means of the respective populations, and σ is SD. Cohen's U_3 was calculated as $U_3 = \varphi(\delta)$, where φ is the cumulative distribution of the normal distribution. Overlap was calculated as the overlapping coefficient OVL using $OVL = 2\varphi(\frac{-|\delta|}{2})$. In all presented analyses, $P < 0.05$ was considered significant and n is defined as each discrete tissue.

Data, Materials, and Software Availability. Data have been deposited in NASA GeneLab public repository (OSD-737, DOI: 10.26030/t9v3-gx23) (88). All other data are included in the manuscript and/or *SI Appendix*.

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